

REALTECH STP

Simple MQTT and SFTP Guide

Updated trip topic, RFID-reader direction rules and exceptional approval flow

Version	Main topic	Theme
Application v0.1.40	Device_Response	Simple, practical and human-friendly

i**What changed**

The trip topic is fixed as IWS_STP-001/Device_Response. Direction comes from the IN or OUT RFID reader. An approved exception follows the same camera, SFTP and MQTT process as a normal trip.

Prepared for operations, server integration, QA and commissioning teams.

1. Required server panel settings

Use the following values for the vehicle trip message. The trip topic is no longer shown as an option between two names.

Setting	Required value	Simple meaning
MQTT Client ID	IWS_STP-001	Identifies this gate device.
Topic name	Device_Response	Use this name for all vehicle trip messages.
Base Topic	IWS_STP-001/Device_Response	Full device trip topic.
Publish Topic	IWS_STP-001/Device_Response	Same full topic used to send trip JSON.
Quality of Service	1	Broker confirms the message was received.
Retain Published Message	Off	Do not keep an old trip as a retained message.



Use only Device_Response

The vehicle trip topic must be IWS_STP-001/Device_Response. Do not use STP_COM in this guide.

Messages at a glance

Purpose	Direction	Topic / Interface	Plain meaning
Vehicle trip	Device -> Server	IWS_STP-001/Device_Response	IN, OUT or approved exception trip.
Recharge request	Server -> Device	IWS_STP-001/STP_RECHARGE_FROM_SERVER	Add balance to one RFID.
Recharge result	Device -> Server	IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE	Applied, Duplicate or Rejected.
Image upload	Device -> SFTP	Configured remote SFTP folder	Upload the image before trip JSON.

2. How IN and OUT direction is identified

The direction is decided automatically by the RFID reader that detects the tag. The server does not need to guess or select the direction.

What happened	direction	What the device does
IN RFID reader detects the tag	IN	Use the IN camera and send an IN trip.
OUT RFID reader detects the tag	OUT	Use the OUT camera and send an OUT trip.



Simple rule

RFID detected by the IN reader means direction is IN. RFID detected by the OUT reader means direction is OUT.

Normal vehicle process



- For a Paid IN trip, the configured entry amount is deducted from the local balance.
- For an OUT trip, the device normally links the earlier IN trip and does not deduct again.
- If SFTP or MQTT is temporarily unavailable, the trip stays pending and is sent later.

3. Normal IN and OUT JSON examples

IN reader example

The IN RFID reader detected the tag, so the JSON direction is IN. The IN camera image is captured and uploaded.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "2663656e-1fa6-4696-b93c-3f46662572aa",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "direction": "IN",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 9800,
    "debitAmount": 100,
    "newBalance": 9700,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...10-33-52 - IN.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...IN.png",
    "processedAt": "2026-07-27T10:33:52.823+05:30"
  }
}
```

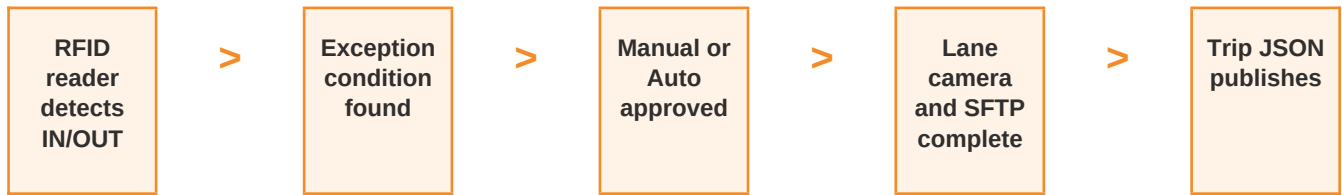
OUT reader example

The OUT RFID reader detected the tag, so the JSON direction is OUT. No second debit is made.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "9a3b5b9e-9c87-420d-abcb-bd2f5f7b8954",
    "direction": "OUT",
    "entryTransactionId": "2663656e-1fa6-4696-b93c-3f46662572aa",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "previousBalance": 9700,
    "debitAmount": 0,
    "newBalance": 9700,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...11-20-44 - OUT.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...OUT.png"
  }
}
```

4. Exceptional approval - same process as IN or OUT

Exceptional approval is used when the normal trip sequence is incomplete, such as a missing IN trip or a missing OUT trip.



Important operation rule

After the exception is approved, the current transaction must continue like a normal IN or OUT trip: capture the image, upload it through SFTP, and send the full trip JSON on Device_Response.

Approval details sent to the server

Manual approval	Auto approval
approvalMode: Manual Name, role and 10-digit mobile are entered by the approver.	approvalMode: Auto Name is SYSTEM, role identifies IN/OUT auto approval, and mobile is N/A.

- reason - why exceptional approval was needed, for example Missing IN Trip or Missing OUT Trip.
- approverName - person name for Manual approval, or SYSTEM for Auto approval.
- approverRole - the person role, or IN AUTO APPROVAL / OUT AUTO APPROVAL.
- approverMobile - approver mobile number for Manual approval, or N/A for Auto approval.
- approvalMode - Manual or Auto, so the server can identify how the approval happened.

5. Exceptional approval JSON examples

Manual approval - OUT reader

The OUT reader detected the RFID. The OUT camera image is captured and uploaded exactly like a normal OUT trip. Approval details are added to the same JSON.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "46bc7046-530c-4242-9913-87b2d19f61d0",
    "direction": "OUT",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "previousBalance": 10000,
    "debitAmount": 0,
    "newBalance": 10000,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...12-10-05 - OUT.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...OUT.png",
    "exceptionalApproval": {
      "approvalMode": "Manual",
      "reason": "Missing IN Trip",
      "approverName": "Arun Kumar",
      "approverRole": "Shift Supervisor",
      "approverMobile": "9876543210",
      "approvalDate": "2026-07-27",
      "approvalTime": "12:09:58.900",
      "timeZoneOffset": "+05:30"
    }
  }
}
```

Auto approval identification

```
"exceptionalApproval": {
  "approvalMode": "Auto",
  "reason": "Missing OUT Trip",
  "approverName": "SYSTEM",
  "approverRole": "IN AUTO APPROVAL",
  "approverMobile": "N/A",
  "approvalDate": "2026-07-27",
  "approvalTime": "12:30:10.200",
  "timeZoneOffset": "+05:30"
}
```

6. Recharge request and response

Server sends recharge request

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_FROM_SERVER",
  "qos": 1,
  "payload": {
    "messageType": "rfidRecharge",
    "rechargeId": "RCH-20260727-000001",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS_STP_001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "previousBalance": 9700,
    "rechargeAmount": 300,
    "newBalance": 10000
  }
}
```

Device sends applied response

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE",
  "qos": 1,
  "payload": {
    "messageType": "rfidRechargeAck",
    "rechargeId": "RCH-20260727-000001",
    "status": "Applied",
    "success": true,
    "duplicate": false,
    "rechargeAmount": 300,
    "previousBalance": 9700,
    "newBalance": 10000,
    "serverBalanceMatched": true,
    "message": "Recharge applied to the local RFID balance."
  }
}
```



Recharge result meaning

Applied = completed. Duplicate = already handled, do not add again. Rejected = not applied; show the returned reason.

7. SFTP upload and final checklist

SFTP moves the captured IN, OUT or exceptional approval image from the device computer to the server folder. MQTT then carries the image path inside the trip JSON.



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Same upload rule for approved exceptions

A Manual or Auto exceptional approval uses the same image capture and SFTP upload process as the current IN or OUT transaction.

Practical checklist

- 1 Set Base Topic and Publish Topic to IWS_STP-001/Device_Response.
- 2 Use QoS 1 and keep Retain disabled.
- 3 IN RFID reader detection must produce direction IN.
- 4 OUT RFID reader detection must produce direction OUT.
- 5 Normal and exceptional approved trips must capture the same-lane camera image.
- 6 Upload the image through SFTP before publishing the trip JSON.
- 7 Include imageFileName and imageRemotePath in Device_Response.
- 8 Include exceptionalApproval only when Manual or Auto approval was used.
- 9 Send approvalMode, reason, approverName, approverRole and approverMobile.
- 10 Store transactionId once to prevent duplicate trip records.
- 11 Reuse the same rechargeId when retrying the same recharge.

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Final understanding

The RFID reader identifies IN or OUT. The camera and SFTP process stays the same for normal and approved exception trips. MQTT sends everything through IWS_STP-001/Device_Response.